Ahmed Mohammed

Applied AI / Machine Learning Engineer



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Born 09.05.1995 in Cairo, Egypt | Driving licence B

DEGREE	THESIS	WORK PERMIT	START
M.Sc. AI, JKU Linz, 2026	**Graded 1 | Prof. Hochreiter**	**RWR Card, no labour market test**	**Available now**

PROFILE

Machine learning engineer with an **M.Sc. in Artificial Intelligence from JKU Linz** (September 2026). My thesis with Univ.-Prof. Dr. Sepp Hochreiter, **graded 1**, uses conditional diffusion models as generative classifiers for out-of-distribution detection: a separation loss I introduced cut the seed-to-seed standard deviation on **CIFAR-10** from 11.07 to 0.07 points and raised mean AUROC to **99.03%**. Strong in Python, PyTorch and computer vision; I use AI-assisted development (Claude Code, GitHub Copilot) openly and daily.

EXPERIENCE

04/2024 – 11/2024

Master's Thesis Researcher (AI/ML)

PROFACTOR GmbH | Steyr, Austria (home office)

- Project **Zer0P** (Government of Upper Austria, zero-defect inkjet printing on building components): built a two-stage **YOLOv8** then conditional-diffusion pipeline for classifying experimental prints produced in a parameter-search feedback loop.
- Annotated the project images myself in Roboflow and trained the vision models in PyTorch Lightning with Hydra configs, so every run is reproducible.

04/2024 – 07/2026

M.Sc. Thesis Research, Machine Vision (part of the degree)

JKU Institute for Machine Learning | Linz, Austria

- Introduced a **class-conditional separation loss** for conditional diffusion models. On **CIFAR-10** over three seeds it cut the seed-to-seed standard deviation from 11.07 to 0.07 points and raised mean OOD detection AUROC from 92.52% to **99.03%**; across five external benchmarks (CIFAR-100, Places365, FashionMNIST, SVHN, Textures) it reached 90.50% to 96.97% AUROC.
- Public code: github.com/ahmed-3m/DiffusionOOD and github.com/ahmed-3m/InkjetOOD.
- Ran an eight-method comparison with five-fold cross-validation under one evaluation protocol on the inkjet data, and reported where a supervised ResNet baseline stayed stronger on small, fine-grained inspection data.

05/2026 – present

Solo Developer and Project Lead, Sihem (personal side project)

Accepted into the tech2b incubator | Linz, Upper Austria

- Design and run the full cloud stack behind an assistant for habits and daily routines, delivered as a Telegram bot and an installable web app: managed PostgreSQL with row-level security, serverless functions in Deno/TypeScript, scheduled jobs, secrets in a managed vault, and deployment through **GitHub Actions**.

08/2023 – 10/2023

AI Research Intern

Karunya Institute of Technology and Sciences | Coimbatore, India

- Built deep learning models for **EEG-based motor imagery classification** in brain-computer interface systems: a decoding problem on noisy, non-stationary, multi-channel time series.

09/2018 – 10/2020

Electronics Technician

Ledzone | Karaköy, Istanbul, Turkey

- Built Internet of Things (IoT) prototypes on Arduino-based microcontroller boards: wired sensors and actuators, programmed the boards, and tested them on the bench.

SKILLS

Deep Learning	PyTorch, PyTorch Lightning, diffusion models (DDPM, UNet), CNNs, Transformers, LSTM
Computer Vision	OOD detection, object detection (YOLOv8), defect inspection, image classification, Roboflow
Research and MLOps	Hydra pipelines, Hugging Face Diffusers, Weights & Biases, ablation studies, AUROC and FPR95, cross-validation, multi-seed runs
LLMs and Cloud	LLM API integration, multi-provider routing with failover, prompt engineering, PostgreSQL with pgvector, serverless functions, GitHub Actions CI/CD
Software	Python (advanced), Git, Linux, Docker, REST APIs; TypeScript and React familiar, built with AI-assisted tooling; Claude Code (daily) and GitHub Copilot

EDUCATION

10/2020 – 09/2026

M.Sc. Artificial Intelligence

Johannes Kepler University (JKU) Linz | Linz, Austria

Graduated September 2026. Thesis: Conditional Diffusion Models as Generative Classifiers for Out-of-Distribution Detection in Inkjet Print Quality Control, **graded 1**. Supervisor Univ.-Prof. Dr. Sepp Hochreiter, assistant supervisor Claus Hofmann, MSc.

02/2015 – 01/2018

B.Sc. Mechatronics Engineering

Eastern Mediterranean University | Famagusta, Cyprus

Graduation project: SCARA robotic system for dynamic object tracking (kinematics, sensor integration, real-time control loop).

10/2011 – 01/2015

Electrical Engineering studies (no degree)

Higher Technological Institute | Cairo, Egypt

Transferred into the Mechatronics Engineering programme at Eastern Mediterranean University. Summers: control panels from schematics in the family electrical business.

LANGUAGES

Arabic	English	German	Turkish
Native	B2 (CEFR)	B1 (CEFR, ÖIF certified), actively improving	Conversational (everyday use)

WORK AUTHORIZATION AND AVAILABILITY

Resident in Linz. I graduated from JKU Linz in September 2026 and am eligible now for the **Rot-Weiß-Rot Karte für Studienabsolventen** (Red-White-Red Card for graduates): **no AMS labour market test**, **no quota**, and no agency or sponsorship cost for the employer, who only confirms the position and salary. **Available now**.

REFERENCES

Univ.-Prof. Dr. Sepp Hochreiter (thesis supervisor, JKU Linz) and Dr. Christian Eitzinger (Head of Machine Vision, PROFACTOR GmbH). Contact details available on request.